Explanation, Features and Development of Personal Portfolio Website

Student Name: Ekansh | **Course & Branch:** B.Tech – CSM (1st Semester) | **College:** DR. RVR NRI Institute of Technology, Vijayawada

1. Introduction

The given code is used to develop a **Personal Portfolio Website** for **Ekansh**, a first-year B.Tech student specializing in Computer Science and Machine Learning (**CSM**) at **DR. RVR NRI Institute of Technology, Vijayawada, Andhra Pradesh**.

The website is created using **HTML** and **CSS**. It presents the student's personal introduction, current learning status, technical skills, project showcase, educational background and profile table, personal interests beyond coding, and career vision in an organized and professional manner.

The website uses a clean light-slate background (`#f4f6f9`) with deep navy (`#234b78`) and royal blue (`#315f91`) highlights, making it visually appealing and easy to read. It uses flexible box layouts so the content adapts smoothly across desktops, laptops, tablets, and mobile screens.

2. Explanation of the Code

The website is developed using two main front-end web technologies: **HTML** and **CSS**.

A. HTML (HyperText Markup Language)

HTML is used to create the structure and semantic content of the webpage. It defines the header, navigation links, hero banner, numbered content sections, tables, lists, and footer. The main HTML elements used in `01-portfolio.html` are:

HTML Element	Purpose in Ekansh's Portfolio Website
`<!DOCTYPE html>`	Declares the document as an HTML5 webpage.
`<html>`	Defines the root container of the HTML document.
`<head>`	Contains page metadata, the document title, and the internal `<style>` sheet.
`<title>`	Sets the browser tab title to *"Ekansh | Personal Portfolio"*.
`<body>`	Contains all visible webpage content rendered in the browser window.
`<header>`	Displays the top brand title (`EKANSH.DEV`) and subtitle (`MY PERSONAL PORTFOLIO`).
`<nav>`	Provides the navy navigation bar with links to `ABOUT`, `SKILLS`, `PROJECTS`, `EDUCATION`, and `INTERESTS`.
`<div>`	Groups elements for layout and styling (`.hero`, `.hero-content`, `.hero-card`, `.container`, `.skills`, `.projects`).
`<section>`	Organizes the portfolio into six distinct, bordered content sections (`01. About Me` to `06. My Vision`).
`<h1>, <h2>, <h3>`	Displays headings of different hierarchy levels across the hero banner, sections, and cards.
`<p>, <span>`	Displays paragraphs, taglines, and accented inline text (such as `Future Developer.`).
`<a>`	Creates clickable anchor links in the navigation bar to jump directly to page sections.
`<table>, <tr>, <th>, <td>`	Creates the structured two-column table in the `04. Education & Profile` section.
`<ul>, <li>`	Displays the bulleted list of personal interests in the `05. Beyond Coding` section.
`<footer>`	Displays the closing motto (`LEARN. CREATE. GROW.`) and author credits at the bottom

B. CSS (Cascading Style Sheets)

CSS is used to control the visual presentation, color palette, typography, card spacing, and responsive Flexbox behavior of the portfolio. The important CSS properties used in `01-portfolio.html` are:

CSS Property	Purpose in Ekansh's Portfolio Website
`background-color`	Sets background colors for the page (`#f4f6f9`), nav bar (`#234b78`), hero (`#e8eef6`), cards (`#ffffff`), and footer (`#1e3552`).
`color`	Defines text colors for body copy (`#263247`), headings (`#234b78`, `#315f91`), and muted labels (`#64748b`).
`font-family`	Sets a clean, readable sans-serif typeface (`Arial, sans-serif`) across the entire document.
`margin & padding`	Controls outer spacing (such as centering `.container` with `margin: auto`) and inner breathing room inside cards and sections.
`display: flex`	Aligns elements in flexible rows inside `header`, `.hero`, `.skills`, and `.projects`.
`flex-wrap: wrap`	Allows flex items (hero panels, skill cards, project cards) to wrap onto new lines automatically on smaller screens.
`gap`	Adds consistent spacing between flexible cards (`35px` in the hero section and `20px` between skill/project cards).
`flex & min-width`	Controls proportional card sizing (`flex: 2` and `flex: 1`) and minimum widths (`140px` to `250px`) before wrapping occurs.
`border & border-radius`	Adds subtle borders (`1px solid #e0e6ee`) and rounded corners (`8px` and `10px`) to sections and cards.
`border-collapse: collapse`	Combines adjacent table borders into clean single lines in the Education & Profile table.
`line-height & letter-spacing`	Improves paragraph readability (`line-height: 1.8` to `2.2`) and adds modern spacing to titles (`2px` to `3px`).
`nav a:hover`	Changes navigation link text color to light blue (`#cbdff7`) when the user hovers over a menu item.

3. Features of the Website

1. Header, Navigation & Hero Section

Introduces the brand header (`EKANSH.DEV — MY PERSONAL PORTFOLIO`), a 5-link navigation bar, and a two-column Hero banner. The left column displays *"HELLO, WELCOME TO MY WORLD — I'm Ekansh. Future Developer."* along with a short bio, while the right column features a `CURRENT STATUS: LEARNING` card highlighting
Python, HTML, and CSS
(*"Learn today. Build tomorrow."*).

2. 01. About Me Section (#about)

Describes Ekansh's background as a first-year B.Tech student with a strong interest in coding, website design, and technology, emphasizing how consistent practice and curiosity drive skill improvement.

3. 02. Technical Skills Section (#skills)

Displays three equal-width responsive cards showcasing core technical competencies:
PYTHON
(*Programming fundamentals*),
HTML
(*Webpage structure*), and
CSS
(*Webpage styling*).

4. Development of the Website

The Personal Portfolio Website is developed through the following systematic steps:

Step 1: Create the HTML file — Create a file named `01-portfolio.html`. Add the standard HTML5 document structure (`<!DOCTYPE html>`, `<html>`, `<head>`, `<body>`) and build the header, navigation bar, hero section, six main content sections (`#about`, `#skills`, `#projects`, `#education`, `#interests`, `.goal`), and footer.

Step 2: Add CSS styling — Inside the `<head>` section, add a `<style>` block defining the page background (`#f4f6f9`), font family (`Arial, sans-serif`), brand colors (`#234b78`, `#315f91`, `#1e3552`), padding, borders, and rounded card corners.

Step 3: Design the sections and cards — Use CSS Flexbox (`display: flex`) to align the header branding, the two-column hero banner (`.hero-content` and `.hero-card`), the 3-card Technical Skills row (`.skills`), and the 3-card Project Showcase row (`.projects`).

Step 4: Build the Education table and Interests list — Create a full-width bordered `<table>` with `border-collapse: collapse` for the academic profile details and an unordered list (`<ul>`) with `line-height: 2.2` for the Beyond Coding section.

Step 5: Make the website responsive — Apply `flex-wrap: wrap`, percentage container widths (`width: 84%`), and `min-width` rules on cards so that columns automatically wrap into a single vertical stack on narrow mobile screens.

Step 6: Test the website — Open `01-portfolio.html` in any web browser (Chrome, Edge, or Firefox). Verify section navigation links, hover effects, table borders, and card alignment across desktop and mobile window widths.

5. Website Folder Structure

Keep your project files organized in your local folder and GitHub repository as shown below:

```
Ekansh-Portfolio/ ├── 01-portfolio.html # Personal Portfolio webpage structure and internal CSS styling ├── 02-college.html # Project 02: DR. RVR NRI Institute of Technology College Website ├── 03-student.html # Project 03: Student Academic Dashboard (Roll No: 26001039) ├── README.md # GitHub repository overview and project documentation └── Documentation.pdf # Complete explanation, features, output, and source code PDF
```

6. Technologies Used

Technology	Usage
HTML5	Creates the semantic structure, headings, sections, links, lists, and profile table.
CSS3	Styles the webpage colors, typography, borders, spacing, and visual hierarchy.
CSS Flexbox	Aligns the header, hero banner, technical skill cards, and project showcase cards.
HTML Tables & Lists	Organizes the Education & Profile data and the Beyond Coding interests list.
Responsive Flex-Wrap	Adapts multi-column card layouts automatically to desktop, tablet, and mobile screens.
Web Browser	Renders and runs the webpage directly without requiring any external server.

The website is a self-contained static front-end webpage. It does not require JavaScript, external CSS libraries, or a backend database, allowing it to load instantaneously in any modern web browser.

7. Advantages of the Website

1. Presents Ekansh's academic and technical profile in a clean, professional format.
2. Showcases foundational skills (Python, HTML, CSS) and first-year web development projects.
3. Provides quick single-click navigation across all major portfolio sections.
4. Displays educational metadata clearly using a structured, easy-to-read HTML table.
5. Uses a cohesive navy and slate-blue color palette with high contrast and readability.

9. Output

Below is the complete rendered browser output of Ekansh's Personal Portfolio Website (`01-portfolio.html`) showing all sections from the top header to the bottom footer:

EKANSH.DEV

MY PERSONAL PORTFOLIO

ABOUT SKILLS PROJECTS EDUCATION INTERESTS

HELLO, WELCOME TO MY WORLD

I'm Ekansh.

Future Developer.

I am a first-year B.Tech student exploring programming and web development. I enjoy learning new skills and transforming ideas into creative digital experiences.

CURRENT STATUS

LEARNING

Python
HTML
CSS

Learn today. Build tomorrow.

01. About Me

I am Ekansh, a first-year B.Tech student with an interest in coding, website design, and technology. I believe that consistent practice and curiosity help me improve my skills. This portfolio presents my learning journey and projects.

02. Technical Skills

PYTHON	HTML	CSS
Programming fundamentals	Webpage structure	Webpage styling

03. Project Showcase

PROJECT 01
Personal Portfolio
A personal website presenting my profile, skills, interests, and career goals.

PROJECT 02
College Website
A webpage organizing college information using HTML elements and CSS styling.

PROJECT 03
Student Webpage
A student information webpage using headings, paragraphs, lists, and tables.

04. Education & Profile

DETAIL	INFORMATION
Name	Ekansh
Course	B.Tech
Branch	CSM
Semester	1st Semester
College	DR. RVR NRI INSTITUTE OF TECHNOLOGY
Location	Vijayawada, Andhra Pradesh

05. Beyond Coding

- Learning programming and problem solving
- Designing and developing websites
- Listening to music
- Watching movies
- Exploring new technologies

06. My Vision

10. Code (Part 1 — HTML Head & CSS Styling)

Complete source code of `01-portfolio.html` developed by Ekansh:

```
<!DOCTYPE html>
<html>
<head>
    <title>Ekansh | Personal
Portfolio</title>

    <style>
        body {
            margin: 0;
            font-family: Arial, sans-serif;
            background-color: #f4f6f9;
            color: #263247;
        }

        header {
            padding: 25px 8%;
            background-color: #ffffff;
            border-bottom: 1px solid
#dce2eb;
            display: flex;
            justify-content: space-between;
            align-items: center;
            flex-wrap: wrap;
        }

        header h2 {
            color: #234b78;
            letter-spacing: 2px;
        }

        header p {
            color: #64748b;
        }

        nav {
            background-color: #234b78;
            padding: 18px;
            text-align: center;
        }

        nav a {
            color: white;
            text-decoration: none;
            margin: 10px 15px;
            font-size: 14px;
            font-weight: bold;
        }

        nav a:hover {
            color: #cbdff7;
        }

        .hero {
            display: flex;
            align-items: center;
            justify-content: space-between;
            gap: 35px;
            flex-wrap: wrap;
            padding: 75px 8%;
            background-color: #e8eef6;
        }

        .hero-content {
            flex: 2;
            min-width: 250px;
        }

        .hero-card {
            flex: 1;
            min-width: 220px;
            padding: 30px;
            text-align: center;
            background-color: #ffffff;
            border: 1px solid #d5deea;
            border-radius: 10px;
        }

        .tag {
            color: #557da8;
            font-weight: bold;
            letter-spacing: 2px;
        }

        .hero h1 {
            font-size: 46px;
            margin: 15px 0;
            color: #1e293b;
        }

        .hero h1 span {
            color: #315f91;
        }

        .hero-content p {
            color: #526176;
            line-height: 1.9;
            font-size: 17px;
        }

        .hero-card h2 {
            font-size: 28px;
            color: #315f91;
        }

        .hero-card p {
            color: #526176;
            line-height: 1.8;
        }

        .container {
            width: 84%;
            margin: auto;
        }

        section {
            margin: 30px 0;
            padding: 30px;
            background-color: #ffffff;
            border: 1px solid #e0e6ee;
            border-radius: 10px;
        }
```

10. Code (Part 2 — Remaining CSS & Complete HTML Body)

```
        .section-title {
            color: #234b78;
            font-size: 25px;
            margin-top: 0;
        }
        .about-text {
            color: #526176;
            line-height: 1.9;
        }
        .skills, .projects {
            display: flex;
            flex-wrap: wrap;
            gap: 20px;
        }
        .skill {
            flex: 1;
            min-width: 140px;
            padding: 25px 15px;
            text-align: center;
            background-color: #f3f6fa;
            border: 1px solid #dce4ef;
            border-radius: 8px;
        }
        .skill h3 { color: #315f91; }
        .skill p { color: #64748b; }
        .project {
            flex: 1;
            min-width: 190px;
            padding: 25px;
            background-color: #f8fafc;
            border: 1px solid #dce4ef;
            border-radius: 8px;
        }
        .project h3 { color: #315f91; }
        .project p {
            color: #526176;
            line-height: 1.8;
        }
        .project-label {
            color: #7186a1;
            font-size: 13px;
            font-weight: bold;
        }
        table {
            width: 100%;
            border-collapse: collapse;
        }
        th, td {
            border: 1px solid #dce4ef;
            padding: 15px;
            text-align: left;
        }
        th {
            color: #234b78;
            background-color: #eaf0f7;
        }
        td { background-color: #ffffff; }
        ul {
            line-height: 2.2;
            color: #526176;
        }
        .goal {
            background-color: #edf3fa;
            border: 1px solid #d5e1f0;
        }
```

```
<body>
    <header>
        <h2>EKANSH.DEV</h2>
        <p>MY PERSONAL PORTFOLIO</p>
    </header>

    <nav>
        <a href="#about">ABOUT</a>
        <a href="#skills">SKILLS</a>
        <a href="#projects">PROJECTS</a>
        <a href="#education">EDUCATION</a>
        <a href="#interests">INTERESTS</a>
    </nav>

    <div class="hero">
        <div class="hero-content">
            <p class="tag">HELLO, WELCOME TO
MY WORLD</p>
            <h1>I'm Ekansh.<br><span>Future
Developer.</span></h1>
            <p>I am a first-year B.Tech
student exploring programming and web
development. I enjoy learning new skills and
transforming ideas into creative digital
experiences.</p>
        </div>
        <div class="hero-card">
            <p class="tag">CURRENT
STATUS</p>
            <h2>LEARNING</h2>
            <p>Python<br>HTML<br>CSS</p>
            <p>Learn today. Build tomorrow.
</p>
        </div>
    </div>

    <div class="container">
        <section id="about">
            <h2 class="section-title">01.
About Me</h2>
            <p class="about-text">I am
Ekansh, a first-year B.Tech student with an
interest in coding, website design, and
technology. I believe that consistent
practice and curiosity help me improve my
skills. This portfolio presents my learning
journey and projects.</p>
        </section>

        <section id="skills">
            <h2 class="section-title">02.
Technical Skills</h2>
            <div class="skills">
                <div class="skill">
<h3>PYTHON</h3><p>Programming
fundamentals</p></div>
                <div class="skill">
<h3>HTML</h3><p>Webpage structure</p></div>
                <div class="skill">
<h3>CSS</h3><p>Webpage styling</p></div>
            </div>
        </section>

        <section id="projects">
            <h2 class="section-title">03.
```